Nils Froleyks

KU Leuven, Belgium

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RESEARCH INTERESTS

Certification, Artificial Intelligence, Automated Reasoning, Hardware Verification, and Planning.

Using formal proofs as a principled way to bridge Artificial Intelligence and Automated Reasoning at scale.

ACADEMIC EXPERIENCE

Postdoctoral Researcher — October 2025 – Present
KU Leuven, Belgium
with Prof. Bart Bogaerts in the Declarative Languages and Artificial Intelligence section

University Assistant — March 2020 – August 2025
Johannes Kepler University Linz, Austria
Teaching duties included courses on Model Checking, Logic, and Debugging.

EDUCATION

Johannes Kepler University Linz, Linz, Austria — March 2020 – August 2025
Ph.D. in Computer Science with Distinction
Thesis: Deep Integration of SAT Solving and Model Checking
Supervisors: Prof. Armin Biere, Prof. Martina Seidl

Karlsruhe Institute of Technology, Karlsruhe, Germany — 2018 – 2020
M.Sc. in Computer Science — **Best Thesis Award**
Thesis: PASAR – Planning as Satisfiability with Abstraction Refinement
Supervisors: Prof. Peter Sanders

Karlsruhe Institute of Technology, Karlsruhe, Germany — 2013 – 2017
B.Sc. in Computer Science
Thesis: Using an Algorithm Portfolio to Solve Sokoban
Supervisors: Prof. Peter Sanders

INDUSTRIAL COLLABORATION

Intel Corporation — Nov 2023 – Present
Continued project on integrating hardware model checking certification into industrial workflows.

Woodpecker Technologies, Singapore — 2022 – 2023
External consultant. Pre-silicon verification using symbolic formal methods.

CAREER BREAKS

Parental Leave — February – November 2022

AWARDS

CAV Distinguished Paper Award — 2026
38th International Conference on Computer Aided Verification
for *Liveness Proofs for Hardware Model Checking*

Top-Three Finalist, Heinz Zemanek Award — 2026
Austrian Computer Society (OCG) award for Austria's best PhD thesis in computer science

CAV Distinguished Paper Award — 2025
37th International Conference on Computer Aided Verification
for *Introducing Certificates to the Hardware Model Checking Competition*

Third Place in Combinatorial Reconfiguration Challenge — 2022
CoRe Challenge 2022, Virtual

Best Master's Thesis — 2020
Karlsruhe Institute of Technology, Germany

First Prize (Partial Gold Medal) — 2019
The Sparkle Planning Challenge, ICAPS, Berkeley, USA

SCIENTIFIC EVENT ORGANIZATION

Organizer, Hardware Model Checking Competition (HWMCC) 2025 FMCAD, Menlo Park, USA
Organizer, Hardware Model Checking Competition (HWMCC) 2024 FMCAD, Prague, Czech Republic
Organizer, SAT Competition 2021 SAT Conference, Barcelona
Organizer, Hardware Model Checking Competition (HWMCC) 2020 FMCAD, Virtual
Organizer, SAT Competition 2020 SAT Conference, Alghero, Italy

SKILLS

Programming Languages: C++, Python
Languages: German (native), English (fluent)

TEACHING & SUPERVISION

Guest Lecturer in Formal Methods for AI **Winter Semester 2026**
Leiden University, Netherlands

Guest Lecturer in SAT solving **Summer Semester 2026**
KU Leuven, Belgium
Advanced course for master's students in computer science.

Supervised Bachelor's Thesis **2026**
Leiden University, Netherlands
Certifying Constraints from Hardware Designs.

Supervised Bachelor's Thesis **2024**
Johannes Kepler University Linz, Austria
Ternary simulation for IC3 Model Checking.

Lecturer in Debugging **Summer Semester 2024**
Johannes Kepler University Linz, Austria
Advanced course for master's students in computer science and artificial intelligence.

Lecturer in Model Checking **Winter Semester 2024**
Johannes Kepler University Linz, Austria
Advanced course for master's students in computer science and artificial intelligence.

Teaching Assistant in Computer Science and Artificial Intelligence Courses **2020–2024**
Johannes Kepler University Linz, Austria
At both the Master's and Bachelor's levels, hands-on teaching courses include Model Checking, Formal Models, Logic, SAT Solving, Practical Work in AI, and Computational Logics for AI.

Teaching Assistant in Algorithms for Planar Graphs **2018**
Karlsruhe Institute of Technology, Germany

Teaching Assistant in Theoretical Foundations of Informatics **2015–2019**
Karlsruhe Institute of Technology, Germany

SOFTWARE

Certifaiger: a certificate checker adopted by the Hardware Model Checking Competition.

Voiraig: a certifying hardware model checker.

Cerbtora: a word-level certificate checker for hardware model checking.

CaDiCaL: an award-winning state-of-the-art SAT solver.

CadiBack: a high-performance backbone extractor built on CaDiCaL.

CONTACTS FOR REFERENCES

Prof. Bart Bogaerts, KU Leuven, Belgium. `bart.bogaerts@kuleuven.be`

Prof. Armin Biere, University of Freiburg, Germany. `armin.biere@gmail.com`

Prof. Martina Seidl, Johannes Kepler University Linz, Austria. `martina.seidl@jku.at`

PUBLICATIONS

[1] **Nils Froleyks**, Emily Yu, Bart Bogaerts, Armin Biere, and Keijo Heljanko. *Liveness Proofs for Hardware Model Checking.* International Conference on Computer Aided Verification (CAV), 2026.
Distinguished Paper Award

[2] Florian Pollitt, Mathias Fleury, Katalin Fazekas, **Nils Froleyks**, André Schidler, Dominik Schreiber, and Armin Biere. *CaDiCaL 3.0: Proofs, Algorithms, Implied Literals, and Robust Scheduling.* Theory and Applications of Satisfiability Testing (SAT), 2026.

[3] **Nils Froleyks** and Emily Yu. *Hardware Model Checking Certification with Certifaiger and Cerbtora.* International Joint Conference on Automated Reasoning (IJCAR), 2026.

[4] Armin Biere, Katalin Fazekas, Mathias Fleury, **Nils Froleyks**, and Florian Pollitt. *Clausal Equivalence Checking.* Journal manuscript under review.

[5] **Nils Froleyks**, Emily Yu, Armin Biere, and Keijo Heljanko. *Certifying Constraints in Hardware Model Checking.* International Symposium on Formal Methods (FM), 2026.

[6] **Nils Froleyks**, Emily Yu, Mathias Preiner, Armin Biere, and Keijo Heljanko. *Introducing Certificates to the Hardware Model Checking Competition.* International Conference on Computer Aided Verification (CAV), 2025.
Distinguished Paper Award

[7] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleyks**, and Florian Pollitt. *CaDiCaL 2.0.* International Conference on Computer Aided Verification (CAV), 2024.

[8] Armin Biere, Katalin Fazekas, Mathias Fleury, and **Nils Froleyks**. *Clausal Equivalence Sweeping.* Formal Methods in Computer-Aided Design (FMCAD), 2024.

[9] Armin Biere, **Nils Froleyks**, and Mathias Preiner. *Hardware Model Checking Competition 2024.* Formal Methods in Computer-Aided Design (FMCAD), 2024.

[10] Armin Biere, Katalin Fazekas, Mathias Fleury, and **Nils Froleyks**. *Clausal Congruence Closure.* Theory and Applications of Satisfiability Testing (SAT), 2024.

[11] **Nils Froleyks**, Emily Yu, Armin Biere, and Keijo Heljanko. *Certifying Phase Abstraction.* International Joint Conference on Automated Reasoning (IJCAR), 2024.

[12] **Nils Froleyks**, Emily Yu, and Armin Biere. *Ternary Simulation as Abstract Interpretation.* Methoden und Beschreibungssprachen zur Modellierung und Verifikation von Schaltungen und Systemen (MBMV), 2024.

[13] Armin Biere, **Nils Froleyks**, and Wenxi Wang. *CadiBack: Extracting Backbones with CaDiCaL.* Theory and Applications of Satisfiability Testing (SAT), 2023.

[14] Armin Biere, Mathias Fleury, **Nils Froleyks**, and Marijn JH Heule. *The SAT Museum.* Pragmatics of SAT, 2023.

[15] **Nils Froleyks**, Emily Yu, and Armin Biere. *BIG Backbones.* Formal Methods in Computer-Aided Design (FMCAD), 2023.

[16] Emily Yu, **Nils Froleyks**, Armin Biere, and Keijo Heljanko. *Towards Compositional Hardware Model Checking Certification.* Formal Methods in Computer-Aided Design (FMCAD), 2023.

[17] Emily Yu, **Nils Froleyks**, Armin Biere, and Keijo Heljanko. *Stratified Certification for K-Induction.* Formal Methods in Computer-Aided Design (FMCAD), 2022.

[18] **Nils Froleyks** and Armin Biere. *Single Clause Assumption without Activation Literals to Speed-up IC3.* Formal Methods in Computer-Aided Design (FMCAD), 2021.

[19] Tomáš Balyo and **Nils Froleyks**. *AI Assisted Design of Sokoban Puzzles Using Automated Planning.* International Conference on ArtsIT, Interactivity and Game Creation, 2021.

[20] **Nils Froleyks**, Marijn Heule, Markus Iser, Matti Järvisalo, and Martin Suda. *SAT Competition 2020.* Artificial Intelligence, 2021.

[21] **Nils Froleyks**, Tomáš Balyo, and Dominik Schreiber. *PASAR—Planning as Satisfiability with Abstraction Refinement.* International Symposium on Combinatorial Search (SoCS), 2019.

[22] **Nils Froleyks** and Tomáš Balyo. *Using an Algorithm Portfolio to Solve Sokoban.* International Symposium on Combinatorial Search (SoCS), 2017.

TECHNICAL REPORTS AND COMPETITION PROCEEDINGS

[1] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleyks**, Florian Pollitt, and André Schidler. *CaDiCaL, IsaSAT and Kissat Entering the SAT Competition 2025.* SAT Competition 2025.

[2] **Nils Froleyks**, Emily Yu, and Armin Biere. *Challenging Certificates from Model Checking.* SAT Competition 2025.

[3] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleyks**, and Florian Pollitt. *Hardware Equivalence Checking Problems Submitted to the SAT Competition 2024.* SAT Competition 2024.

[4] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleyks**, and Florian Pollitt. *CaDiCaL, Gimsatul, IsaSAT and Kissat Entering the SAT Competition 2024.* SAT Competition 2024.

[5] **Nils Froleyks**, Emily Yu, and Armin Biere. *ReconfAIGERation entering Core Challenge 2022.* CoRe Challenge 2022.

[6] **Nils Froleyks**, Emily Yu, and Armin Biere. *Unique Reconfiguration Sequence.* SAT Competition 2022.

[7] Emily Yu, **Nils Froleyks**, Armin Biere, and Mathias Fleury. *Hardware Model Checking Certificates.* SAT Competition 2022.

[8] Tomáš Balyo, **Nils Froleyks**, Marijn JH Heule, Markus Iser, Matti Järvisalo, and Martin Suda. *Proceedings of SAT Competition 2020: Solver and Benchmark Descriptions.* 2020.

[9] **Nils Froleyks**, Tomáš Balyo, and Dominik Schreiber. *PASAR Entering the Sparkle Planning Challenge 2019.* Sparkle Planning Challenge 2019.